Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2023
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) What is a design pattern? [2]
b) How model and view objects are decoupled in MVC? [3]
c) List the appearance methods of basic glyph interface. [2]
d) Write the intent, scope, and alias name of Abstract Factory pattern. [3]
e) Draw the object diagram for composing a TextView object using Decorator pattern. [2]
f) What are the consequences of applying Proxy design pattern? [3]
g) How Observer design pattern is applied to messaging application? [2]
h) Explain how enhanced modularity is achieved with mediator design pattern? [3]
i) How Mediator and Singleton patterns are related with Observer pattern? [2]
j) What may the specific patterns to be applied for database refactoring depend upon? [3]

PART – B

(50 Marks)

2. Describe design patterns with the help of a structured template. [10]

OR

3.a) Explain the step-by-step approach to use a design pattern effectively.
b) Explain the principle ‘Favor object composition over class inheritance’. [7+3]

4.a) Explain the Formatting algorithm in Document Structure.
b) Compare the Glyph and Mono Glyph. [5+5]

OR

5.a) Write the intent and motivation for Prototype pattern.
b) Explain collaborations and consequences of Builder pattern. [5+5]

6. Explain in detail the composite pattern. [10]

OR

7.a) Write the applicability, collaborations, and consequences of flyweight pattern. [5+5]
b) Write the intent, motivation, and structure of Proxy pattern.

8.a) Write the intent, applicability, and consequences of Chain of Responsibility pattern.
b) Write the intent, alias name, structure and Related patterns of Command pattern. [5+5]

OR

9. Take part in explaining how Mediator design pattern is used in Multiplayer games.[10]

10.a) Write intent, alias name, applicability, and structure of State pattern.
b) Explain in brief how Strategy pattern is used in financial trading systems? [5+5]

OR

11. Write short notes on the following:
a) Document and Learning Aid.
b) Life Cycle of Object-Oriented Software. [5+5]

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